

TrES-3b

R-0398

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180611 · created 2026-07-19T03:27:48+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458280.8072 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.011
Transit depth (Rp/Rs)^2	3.59 +/- 0.4 [%]
Orbital Inclination (inc)	82.55 +/- 0.44
Airmass coefficient 1 (a1)	1.121 +/- 0.012
Airmass coefficient 2 (a2)	-0.11 +/- 0.039
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#2 - [328, 305]
Optimal Aperture	4.81
Optimal Annulus	8.76
Transit Duration (day)	0.0634 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.189 ± 0.011 vs 0.16309 (deviation 2.36σ)
Duration fit vs published	0.0634 d vs 0.0567 d (deviation 1.92σ)
Mid-time O–C	2.2 min
Depth SNR	17.2
Residual scatter	1.02 %

Light curve

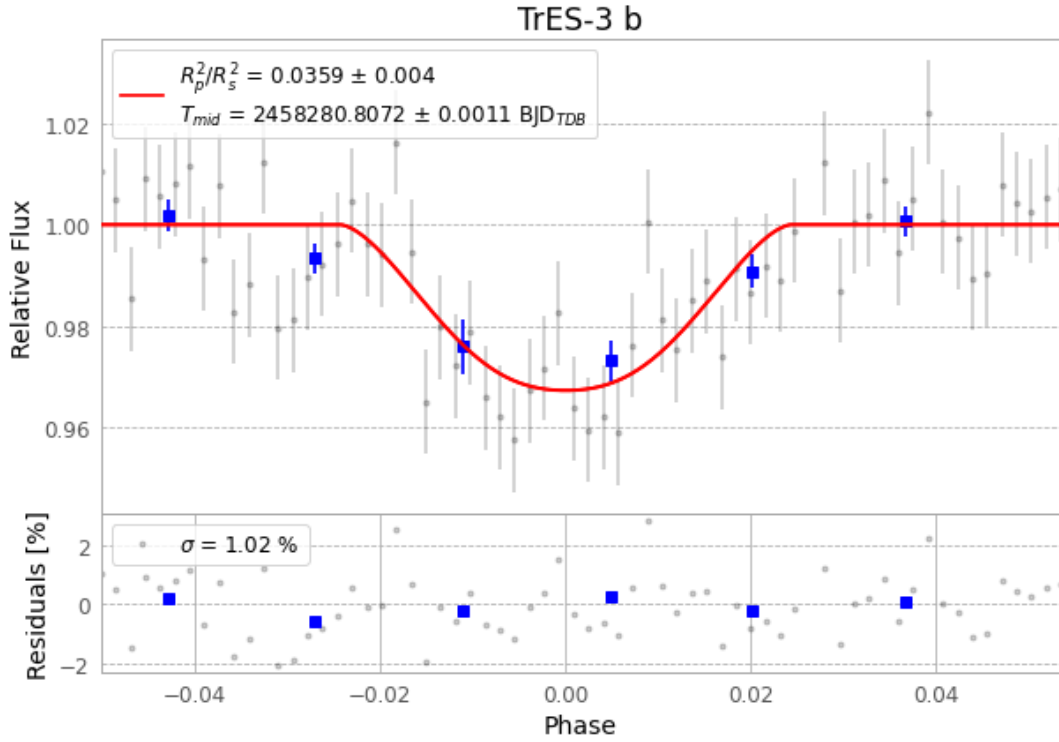


Figure 1 — FinalLightCurve_TrES-3 b_2018-06-10.png (SHA-256 eb8a3df4982a9384...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df9d4fc582ef4cc7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

66 FITS frames (43 MB), TRES-3180611054212.FITS → TRES-3180611085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180611.log (228944efc678...), FinalLightCurve_TrES-3 b_2018-06-10.png (eb8a3df4982a...), FinalLightCurve_TrES-3 b_2018-06-10.csv (3bf4e8ba7f1e...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 03:27Z.